USC BIOLOGICAL MUSEUM · BIOSPHERE · BACKEND

Backend progress report

Every automated backend check on `develop`, a signed-out call to every API route, 70 signed-in API calls as the Curator and role checks as the Developer, all against a local build of `develop` connected to the shared database. Plus a fresh look at the SRS gaps and the defect log since 28 Oct. Only sign-ins and sign-outs were written; nothing was created, edited or emailed.

Date 3–4 Nov 2026 **Repo** biosphere-backend **Commit** develop @ 06545c0 **Previous** 28 Oct @ edaf18c

Roles tested Curator, Developer **Database** hosted Supabase (read-only)

93%	1,146	210	3
of SRS requirements met on the backend, same as 28 Oct	automated tests passed (703 unit, 443 e2e)	API calls (140 signed out, 70 as Curator) all behaved as designed	e2e failures, all from the local `.env`, not the code

Progress

The backend score did not move this week. The 16 merged PRs went into fixes, security patches, a new XLSX import and acceptance evidence, not into the five remaining SRS gaps listed under Modules.

4 Oct	28 Oct	4 Nov (now)
88%	93%	93%

29 OCT – 2 NOV	3 – 8 NOV · NOW	9 – 13 NOV	14 – 18 NOV	19 – 22 NOV	23 NOV
Confirm remaining list, repair fixtures	Finish required features. Feature target 8 Nov	Pinned release candidate and end-to-end demo pass	Last fixes, retests and sign-off. Completion 18 Nov	Rehearsal on the frozen demo	Defense

> Dates come from `docs/DELIVERY_PLAN_2026-11-18.md`. Five days remain before the 8 Nov feature target, and backend `main` was last updated on 10 Sep.

Completion by module

Backend share of each SRS v0.10 section (§4.1 to §4.15), scored the same way as the 28 Oct completion report: 1 for done, 0.5 for partial, 0 for missing. Owners are from the acceptance plan.

Complete Short of 100% Under 50%

§4.1	Authentication and accounts Voltaire Curators cannot activate, deactivate or hand over accounts (Developer-only). No inactivity timeout.	85%
§4.2	Developer functions Voltaire AR asset metadata and compatibility not checked on the server. `/developer/curators` reports pending invitations as Active.	97%
§4.3	Dashboard and notifications Charles, John No notifications for sync failures or other operational events. Read and dismiss stays off until `NOTIFICATION_STATE_ENABLED=true` .	88%
§4.4	Inventory and cataloging Kyer, John	100%
§4.5	Lots and quantities John	100%
§4.6	Storage locations Kyer	100%
§4.7	Search, reports and export John	100%
§4.8	General inquiries Charles, John	100%
§4.9	Visit requests Charles	100%
§4.10	Public website Charles, John, Voltaire	100%
§4.11	FAQ assistant Kyer	100%
§4.12	QR exhibits Voltaire	100%
§4.13	WebAR Voltaire	100%
§4.14	Offline drafts and sync Kyer	100%
§4.15	Audit, backup and recovery Kyer Backup history is read-only. A real backup, restore test and failure alerts wait for an approved procedure and restore target.	44%

> 11 of 15 sections are complete. No backend-only gap closed between 28 Oct and today, so the figures are carried forward. Code-complete is not the same as accepted: every module's sign-off in the acceptance plan is still Pending.

Updates since 28 Oct

16 PRs merged into backend `develop`. Six change the product or its dependencies; the rest are tests and acceptance records.

MERGED	PR	WHAT CHANGED
3 Nov	#223	Visit requests accept several vehicles. A plate listed twice and nested arrays in list fields are rejected
2 Nov	#221	Lot history reads the retained `TEMPORARY_ABSENCE` adjustment instead of failing with a 500 (DEF-030, the high finding from 28 Oct). New write choices stay limited to the original seven
31 Oct	#220	Dependency security patches: CSV parser, Swagger YAML, Prisma patch, Multer for uploads, request parsing and Handlebars
29 Oct	#217	`proxy-addr` security update to 2.0.8
29 Oct	#216	XLSX specimen import through the reviewed preview and commit flow, with a streaming XML check that replaced a slow scan (DEF-029, verified)
29 Oct	#214	Storage-inventory denial audit fixture repaired (DEF-017, now verified)
31 Oct – 2 Nov	#222, #224, #225	Offline boundary test matrix; 30 curator-supplied museum FAQ answers prepared for content approval and evaluated through the public API
28 – 30 Oct	#210 – 213, 215, 218, 219	Documentation: notification activation, public form fixes, module checkpoints, security and XLSX closeouts

Test results

CHECK	RESULT	DETAIL
Lint (`lint:ci`)	PASSING	0 errors, 6 warnings, all from unused `eslint-disable` comments at the top of four test files
Unit tests	PASSING	703 of 703 in 53 suites (701 on 28 Oct)
Build	PASSING	`nest build` succeeds
E2E tests	3 LOCAL FAILURES	440 passed, 17 skipped (the opt-in live Developer suite), 3 failed in the exhibits suite. Rerun with the default public site address: exhibits 77 of 77. See the first issue below
QR timing test	PASSING	"returns a PNG QR code by default", which timed out on 28 Oct, passed in the full run this time. It still has the 5-second default timeout

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API, signed out	WORKING	All 140 routes called against a local build. 130 protected routes answer 401. Public: root and two FAQ reads answer 200, empty inquiry, visit, FAQ ask and FAQ review posts answer 400, an unknown exhibit slug is 404, login without credentials is 401. Onboarding completion was skipped
API as Curator	WORKING	70 read-only calls across every module: dashboard, notifications, specimen search and detail, lots, collections, tags, storage (path, children, inventory, movements, capacity), inquiries, visit requests in all six statuses, exhibits with QR and label, FAQ, audit log, backups, reports and users. All returned the expected result. By design: the FAQ review queue answers 503 because it is off, a Completed visit's campus-entry summary answers 400, and Developer routes answer 403
API as Developer	WORKING	Curator list (8 accounts) and AR exhibit list (34 exhibits) load. All 14 curator routes tried answer 403 "Requires role: CURATOR"
DEF-030 retest	PASSED	`GET /specimens/:id/lots/history` for `USC-BM-2026-0001` returns 200 with both entries, including the retained Temporary absence adjustment, where it returned 500 on 28 Oct
Response time	UNDER 3 S	Signed-in calls: median 0.35 s, 95th percentile 0.71 s, slowest 1.0 s (specimen search). Public reads 0.02 – 0.17 s. On 28 Oct, calls took 1.5 – 8 s from behind a VPN. Measured from this laptop, not the deployment host
Database tests	NOT RUN	They need Docker Desktop for a throwaway Postgres, and it was not running. Includes the lot-history database suite for DEF-030
Anything that writes	NOT TESTED	By choice: create, edit, archive, imports, uploads, replies, status changes, invitations and AR deployment

Issues

LOW

New

Exhibits e2e tests pick up the local public site address

Three exhibits tests expect QR links and labels to use `http://localhost:3000`. On this laptop they received the address set in `.env` as `PUBLIC_SITE_URL`, so they failed. With that variable set back to the default, the suite passes 77 of 77. CI has no `.env`, so it is not affected.

Where	`test/exhibits.e2e-spec.ts`; default in `src/exhibit/exhibit-qr.service.ts:5`
Fix	Set `PUBLIC_SITE_URL` explicitly inside the exhibits e2e setup so the suite doesn't depend on whoever runs it.
Also check	The local value is a temporary Cloudflare tunnel address. QR codes or labels generated by a server with that setting will stop working when the tunnel closes. Make sure the deployed value is the permanent public site before printing labels.

FIXED

Retest passed

Lot history 500 on the shared database (DEF-030)

The high finding from 28 Oct is fixed on `develop`. Signed in as the Curator, the API returns both history entries for `USC-BM-2026-0001` and the Lots page shows "Quantity adjustment · Temporary absence". The defect log still says Ready for retest; record this result there. It was run against a local build, so recheck

defect log still says Ready for retest, record this result there. It was run against a local build, so recheck once the deployment is updated.

LOW

Carried over

Developer curator list shows pending invitations as Active

Confirmed again as the Developer: `GET /developer/curators` returns all 8 accounts with status ACTIVE and no invitation field, so a Developer can't see which curators have never set a password.

LOW

Observation

A visit can be marked Completed before its visit date

Visit 5777AFE9 is Completed although its approved schedule is Wed 11 Nov. The status policy only checks the order of statuses, not the date. It's a test record, but decide whether Completed should wait until the approved date has passed.

Where `src/visit-requests/visit-request-status.policy.ts`

LOW

New

Unused lint suppressions in four test files

Lint reports 6 warnings because four test files start with `eslint-disable` comments for rules that no longer fire. They don't fail CI, but removing them keeps lint output clean.

Where `src/specimen-lots/specimen-lots.service.spec.ts` , `test/audit.e2e-spec.ts` , `test/faq.e2e-spec.ts` , `test/specimen-lots.e2e-spec.ts`

Acceptance defect log

From `docs/TEST_DEFECT_LOG.md` today. 30 defects, up from 26 on 28 Oct. Three more are verified (DEF-017, 028, 029), and four were added (DEF-027 to 030).

STATUS	28 OCT	3 NOV	DEFECTS
VERIFIED	4	7	DEF-002, 009, 017, 018, 025, 028, 029
READY FOR RETEST	19	21	Includes the three majors DEF-001, 003 and 030, plus the new DEF-027
OPEN	2	1	DEF-019 (offline tab-interruption diagnostic wrote to the server)
IN PROGRESS	1	1	DEF-011 (Clear all pending feedback)

Next for the backend

- Record the passing DEF-030 retest in the defect log, then work down the other 20 ready-for-retest defects.
- Get approval for the backup procedure and a disposable restore target, then run one real backup and restore (§4.15, the biggest gap).
- Decide whether curators may activate, deactivate and hand over accounts, or whether that stays Developer-only (§4.1).
- Raise notifications for sync failures and the other events in REQ-4.3-10, and switch on `NOTIFICATION_STATE_ENABLED` in each deployment.

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- Return invitation state from `/developer/curators`, and pin `PUBLIC_SITE_URL` in the exhibits e2e setup.
- Release `develop` to `main` once the retests pass.

Backend progress report for BioSphere, 3–4 November 2026. Earlier reports: System test report, 28 Oct · Completion report, 28 Oct